

Automated Question Bank Generation System

DSA-3 • REVIEW 2

Section: A5

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INTRODUCTION

Automated Question Bank Generation System

An intelligent system that automatically generates structured examination questions from textbooks, lecture notes, and educational resources — extracting keywords, classifying topics, estimating difficulty, and organizing everything into a searchable repository.

- **Reduces manual effort**
- **Consistent question quality**
- **Searchable repository**



Education



Books & Notes



AI / NLP



Question Papers



Database

Prior Work on Automatic Question Generation



PAPER 1

A Systematic Review of Automatic Question Generation for Educational Purposes

SUMMARY

- Reviews AQG techniques
- Covers rule-based, template-based & NLP approaches
- Discusses educational assessment automation

LIMITATIONS

- Limited difficulty estimation
- No complete question bank management
- Inconsistent evaluation methods



PAPER 2

Automatic Question Generation and Answer Assessment: A Survey

SUMMARY

- Reviews automatic question generation
- Explains NLP-based approaches
- Discusses automatic answer evaluation

LIMITATIONS

- Requires manual verification
- Weak organization & retrieval of questions



PAPER 3

Automatic Question Generation Using NLP and Transformers

SUMMARY

- Uses Transformer models
- Generates MCQs and descriptive questions
- Produces higher-quality questions

LIMITATIONS

- High computational cost
- Requires large datasets
- Difficult to deploy on low-resource systems

Where Existing Systems Fall Short

Existing Systems	Research Gap
Rule-based systems	Limited flexibility
NLP-based systems	Weak topic organization
Transformer models	High computational cost
Existing AQG systems	Lack searchable repositories & accurate difficulty estimation



PROPOSED SOLUTION

- ✓ Automatically generate structured question banks
- ✓ Extract important keywords
- ✓ Categorize questions by topic
- ✓ Estimate difficulty levels
- ✓ Store questions in a searchable repository
- ✓ Reduce manual effort for educators

CONCLUSION

From Fragmented Techniques to a Unified Platform

Existing Automatic Question Generation systems mainly focus on generating questions but provide limited support for organization, searchable repositories, and difficulty estimation. Our proposed system integrates these capabilities into a single platform — an efficient, practical solution for educational institutions.



A single pipeline — from raw content to a ready-to-use, searchable question bank.

REFERENCES

Sources & Documentation

RESEARCH PAPERS



A Systematic Review of Automatic Question Generation for Educational Purposes

Springer



Automatic Question Generation and Answer Assessment: A Survey



Automatic Question Generation Using NLP and Transformers

IEEE

TECHNICAL DOCUMENTATION



Python Documentation



Java Documentation



Spring Boot Documentation



spaCy Documentation



NLTK Documentation